

PTTA: Purifying Malicious Samples for Test-Time Model Adaptation

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Introduction

- Test-Time Adaptation (TTA)
 - goes beyond large-scale and heavy pre-training
 - utilizes unlabeled test samples to enhance the online prediction performance of pretrained models
- Since TTA algorithms seek to avoid human supervision and intervention
- adopt self-supervised tasks
 - Entropy Minimization (Wang et al., 2021)
 - Consistency Learning (Wang et al., 2022)
 - models' biases & catastrophic forgetting
- **selecting benign samples**

Introduction

- selecting benign samples
 - filter and discard a portion of test samples
 - data scarcity issues
 - encounter a trade-off
 - stable optimization
 - comprehensive use of test samples

➤ purify malicious samples
by using retrieved benign samples

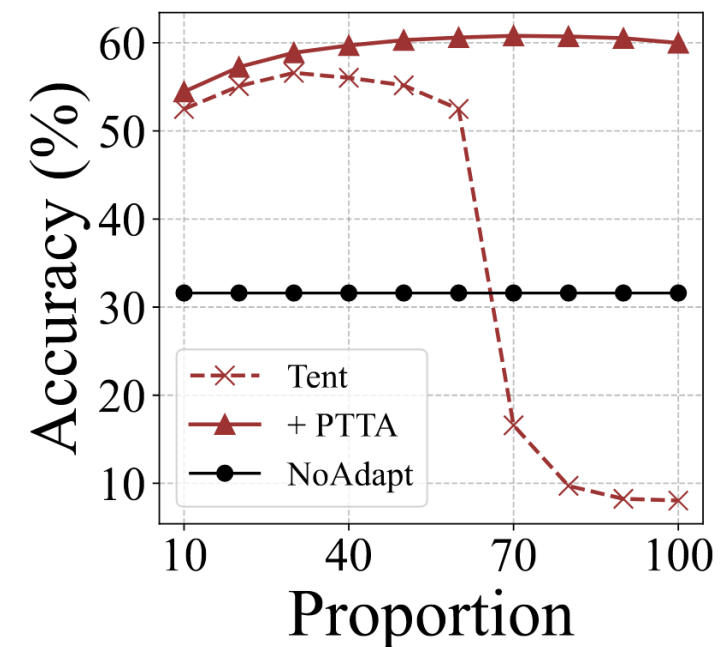
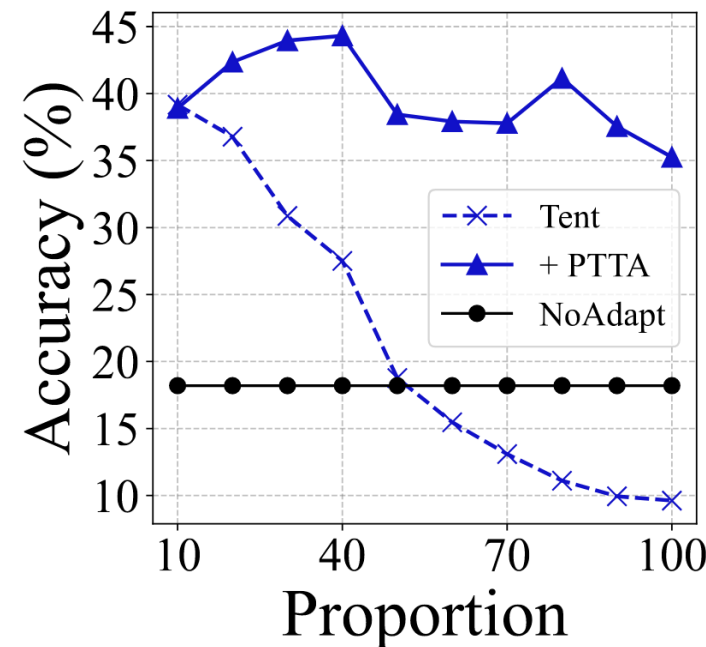
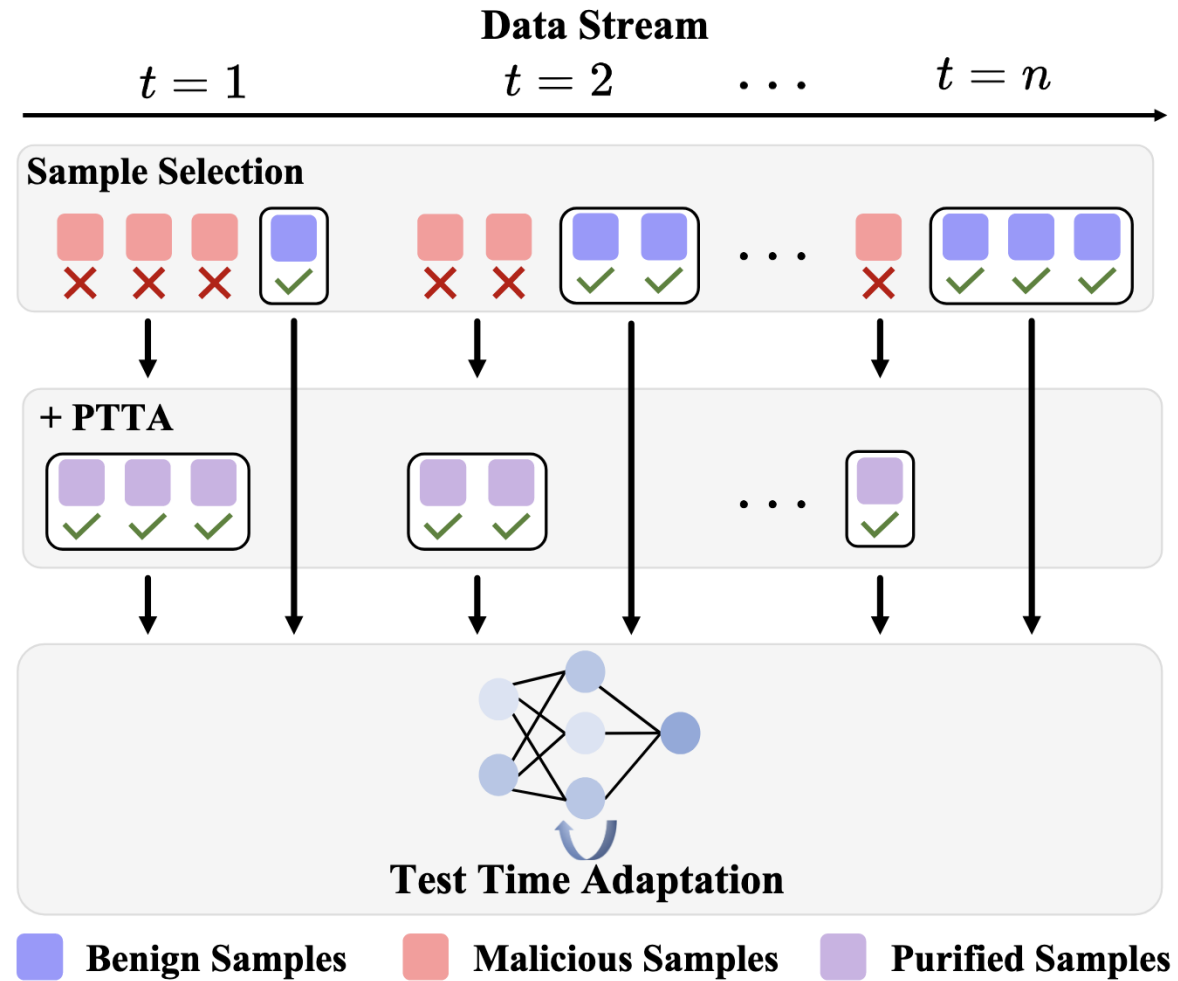


Figure 2: We vary the proportion of selected benign (low-entropy) samples and plot the test accuracy curve of ResNet50 (left) and ViT-B/16 (right).

Overview



Preliminary

- Entropy Minimization

$$\min_{\theta} -\frac{1}{N_{bs}} \sum_{i=1}^{N_{bs}} \sum_{j=1}^C \mathbb{1}_{\{S(x_{ti}) > S_0\}} \cdot p_j(x_{ti}, \theta) \log p_j(x_{ti}, \theta)$$

- Indicator function $\mathbb{I}$: selection mechanism
 - Entropy
 - Cosine similarity
 - PLPD
 - Etc..

Preliminary

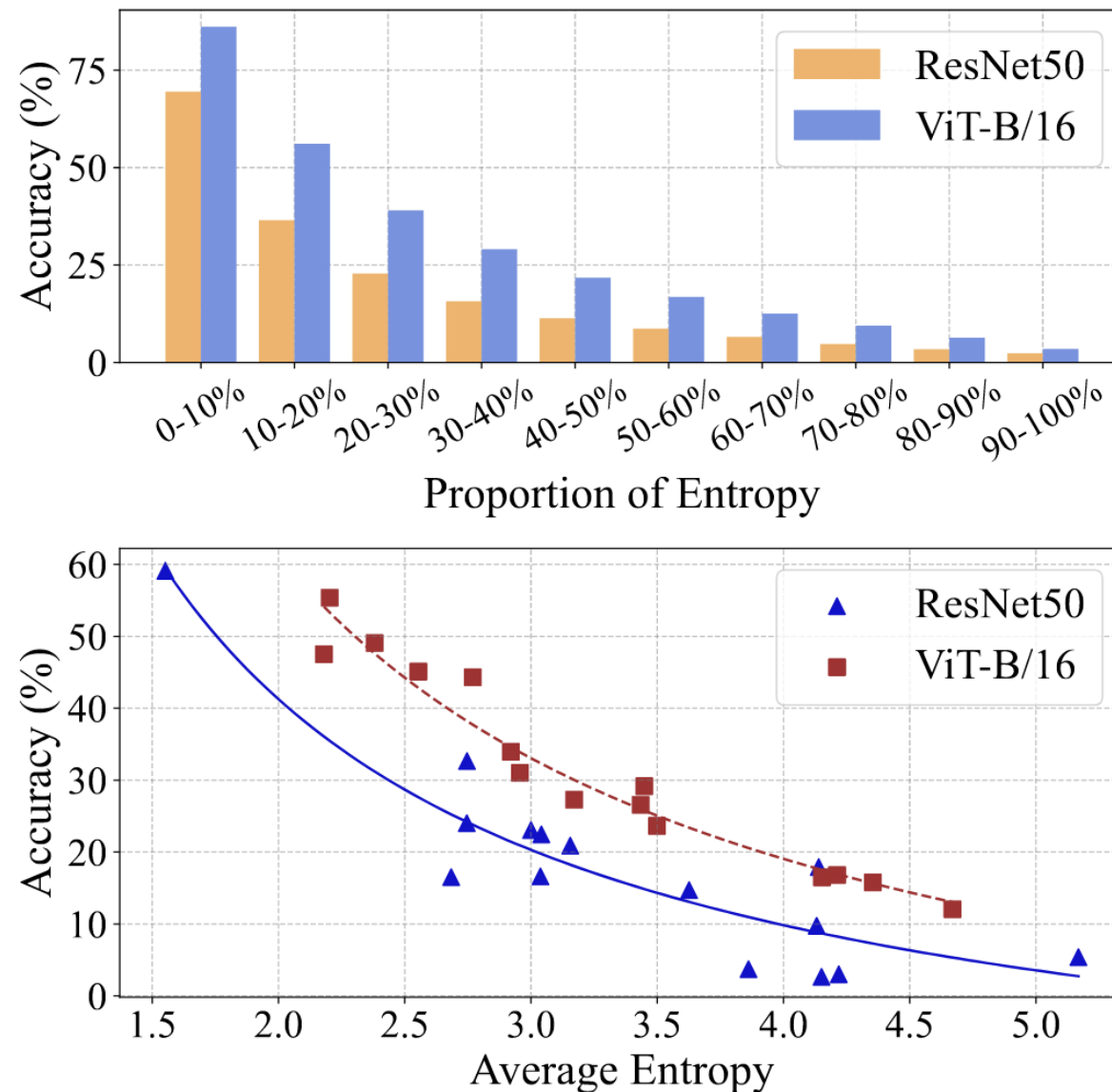


Figure 3: We test the accuracy of top- $p\%$ low-entropy samples (top), and visualize the relationship between the averaged prediction entropy and the test accuracy of DNNs for 15 types of out-of-distribution domains (down).

Pipeline of Methods

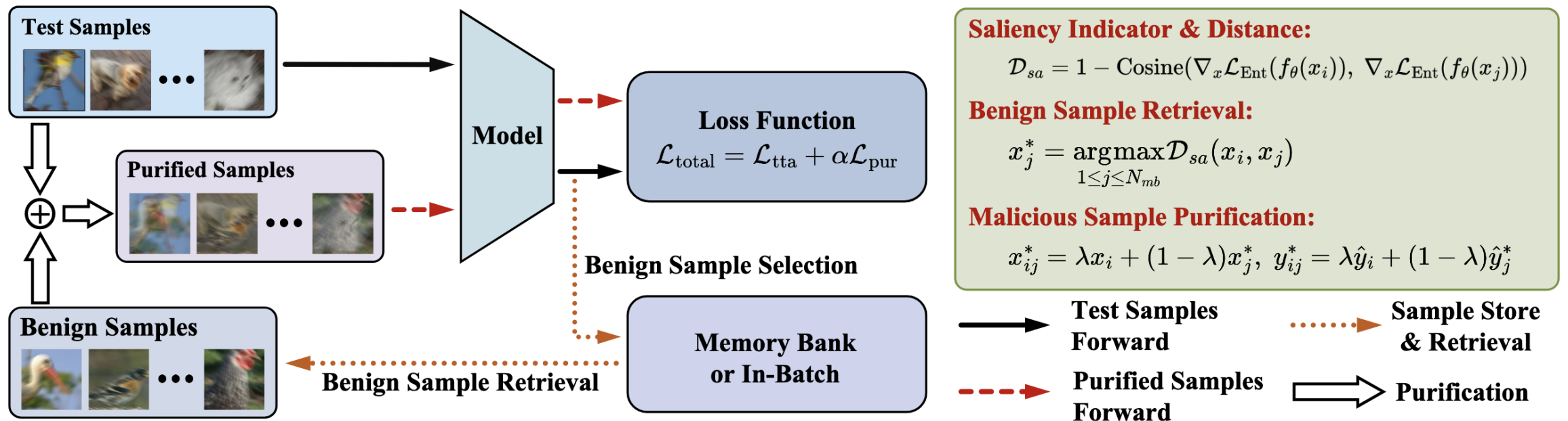


Figure 4: The pipeline of Purification for Test-Time Adaptation (PTTA). We purify malicious test samples using benign samples retrieved from a memory bank or within the mini-batch, based on the saliency indicator we proposed for encoding benign and malicious data. The purified samples are used to calculate the purification loss $\mathcal{L}_{\text{pur}}$, which is directly inserted into the original objective function $\mathcal{L}_{\text{tta}}$ of existing TTA methods.

Methods

- saliency information
 - first-order derivative $\nabla_x L$
 - judge how pixels in the image affect the model's
 - indicates the importance of each pixel
 - measures which individual pixels need to be changed the most in order to minimize the objective function
- samples contributing similarly to the objective function exhibit aligned directions of their saliency information → saliency information of Entropy Minimization: $\nabla_x L_{Ent}(f_\theta(x))$

Methods

- saliency indicator
 - Pixel-saliency indicator: $\nabla_x L_{Ent}$
 - encodes the contribution to objective functions in the high-dimensional sample space, which preserves spatial or temporal information
 - vulnerable to pixels that are irrelevant to the model prediction
 - Feature-saliency indicator: $\nabla_h L_{Ent}$
 - encodes images, it reduces the interference of irrelevant pixels
 - possess semantic information and are more closely related to the model's final decisions
 - They need backprop

Methods

- saliency indicator
 - Logit-saliency indicator: $\nabla_z L_{Ent}$
 - directly encodes the impact of test samples on the model's final decisions
 - can be directly calculated without the need of backpropagating gradients
 - $\nabla_z L_{Ent}(f_\theta(x)) = -(z - p \cdot z) \odot p$

Methods

- saliency indicator

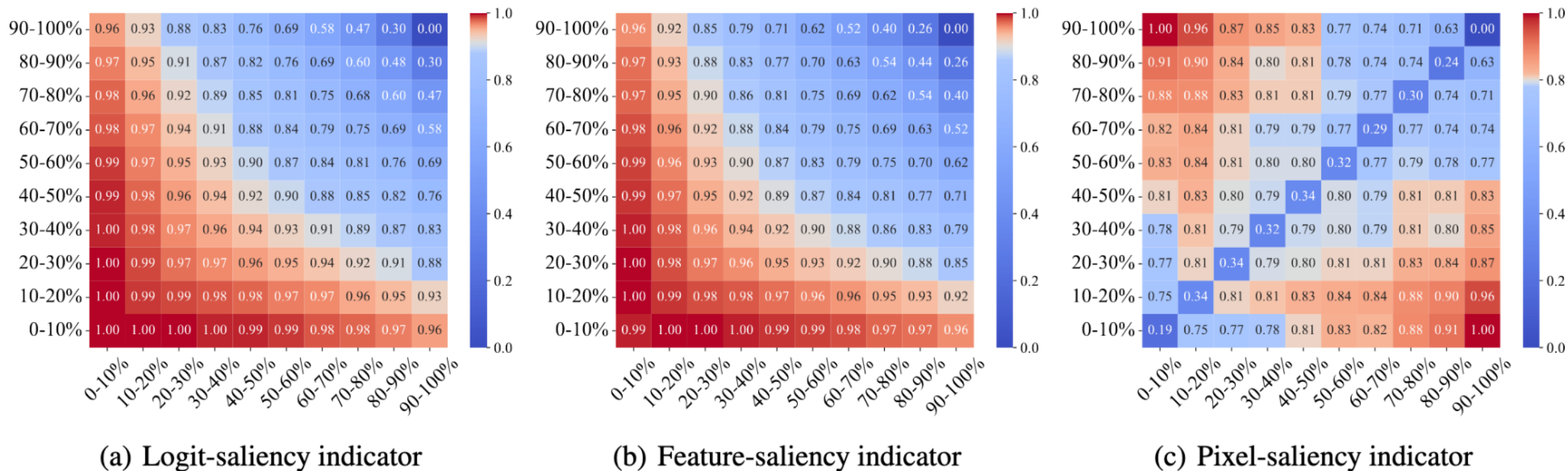


Figure 6: The Saliency Distance (normalized to the range of 0 ~ 1) among test samples sorted in the ascending order of prediction entropy and split by percentages. A good indicator satisfies $\mathcal{D}_{sa}(x^+, x^+) > \mathcal{D}_{sa}(x^+, x^-) > \mathcal{D}_{sa}(x^-, x^-)$.

Methods

- Benign Sample Retrieval
 - gradients corresponding to benign samples x^+ and those corresponding to malicious samples x^- have opposite directions
 - $\text{Cosine}\left(\nabla_z L_{Ent}(f_\theta(x^+)), \nabla_z L_{Ent}(f_\theta(x^-))\right) < 0$
 - $D_{sa} = 1 - \text{Cosine}\left(\nabla_z L_{Ent}(f_\theta(x_i)), \nabla_z L_{Ent}(f_\theta(x_j))\right)$
 - $x_j^* = \arg \max_{1 \leq j \leq N_{mb}} D_{sa}(x_i, x_j)$

Methods

- Benign Sample Retrieval
 - Memory bank

방법	저장소	업데이트 방식	특징
ID Bank	훈련 데이터셋(Training set)의 benign 샘플 전체	사전 구축된 고정 목록(offline)	- 대용량(수십만 샘플) → 검색 비용 높음 - 도메인 차이(훈련↔테스트) 가능성
In-Batch	현재 미니배치 안의 benign 샘플	매 배치마다 mask된(엔트로피 낮은) 샘플만 필터링하여 사용	- 메모리·연산 부담 거의 없음 - 배치 크기에 따라 후보 부족 가능
OOD Bank	테스트 스트림 중 최근에 성공적으로 적응된 benign 샘플 k개	FIFO 큐로 유지: 매 배치에서 saliency 기준을 통과한(benign) 샘플을 추가하고, 오래된 것은 버림	- 도메인 드리프트 반영된 최신 benign 샘플 보유 - 은근한 메모리·검색 비용(기본 k=1 000)

Methods

- Malicious Sample Purification

- Using mixup

- $x_{ij}^* = \lambda x_i + (1 - \lambda)x_j^*$

- $y_{ij}^* = \lambda \hat{y}_i + (1 - \lambda)\hat{y}_j^*$

- $\lambda = 1/(K + 1)$, K decides that the top-K samples with the largest saliency distance are retrieved from the scope

- $$\mathcal{L}_{\text{pur}} = -\frac{1}{N_{bs}} \sum_i^{N_{bs}} \sum_c^C (y_{ij}^*)_c \log \sigma_c(f_\theta(x_{ij}^*)).$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{tta}} + \alpha \mathcal{L}_{\text{pur}},$$

Experiments

Table 1: We demonstrate the generality of PT TA across various sample selection criteria. Meanwhile, we compare the performance of PT TA with different saliency indicators, types of retrieval, and sample purification scopes. We count and compare the number of forward (#Fwds) and backward (#Bwds) propagations for different methods.

METHODS	SINGLE				CONTINUAL				#FWDS	#BWDS
	ETA	DEYO	CPL	AVG.	ETA	DEYO	CPL	AVG.		
COMPARISON WITH EXISTING PURIFICATION (PURIF.) METHODS.										
BASE ALGORITHM	48.2±0.04	48.6±0.11	42.8±0.04	46.5	62.2±0.07	62.2±0.35	55.5±0.13	60.0	N_{batch}	N_{batch}
+ PURIF. W/ FGSM	45.9±0.04	47.0±0.04	42.1±0.09	45.0	61.7±0.03	62.2±0.59	56.4±0.08	60.1	$N_{batch} \times 2$	$N_{batch} \times 2$
+ PURIF. W/ SOAP (5 ITER)	47.4±1.07	47.7±0.25	36.6±8.80	43.9	57.0±7.43	64.1±0.09	49.1±3.22	56.7	$N_{batch} \times 6$	$N_{batch} \times 6$
ABLATION STUDY OF DIFFERENT SALIENCY INDICATORS (IND.).										
+ PIXEL-SALI. IND.	49.0±0.03	50.0 ±0.09	43.0±0.06	47.3	65.6 ±0.02	66.0 ±0.01	<u>59.3</u> ±0.05	63.6	$N_{batch} \times 2$	$N_{batch} \times 2$
+ FEATURE-SALI. IND.	<u>49.1</u> ±0.04	49.9±0.03	<u>43.4</u> ±0.06	<u>47.5</u>	<u>65.4</u> ±0.05	<u>65.8</u> ±0.02	58.9±0.06	63.3	$N_{batch} \times 2$	$N_{batch} \times 2$
+ LOGIT-SALI. IND.	49.5 ±0.00	<u>49.9</u> ±0.10	44.6 ±0.06	48.0	65.3±0.06	65.8±0.01	59.6 ±0.01	<u>63.5</u>	$N_{batch} \times 2$	N_{batch}
ABLATION STUDY OF DIFFERENT TYPES OF RETRIEVAL (RET.).										
+ ID RET. ($N_{mb} = 1k$)	48.2±0.25	48.3±0.22	43.1±0.18	46.5	64.4±0.10	63.3±0.49	58.2±0.08	62.0	$N_{batch} \times 2$	N_{batch}
+ OOD RET. ($N_{mb} = 1k$)	49.5 ±0.00	<u>49.9</u> ±0.10	44.6 ±0.06	48.0	<u>65.3</u> ±0.06	<u>65.8</u> ±0.01	59.6 ±0.01	63.5	$N_{batch} \times 2$	N_{batch}
+ IB RET. ($N_{mb} = 0$)	<u>49.3</u> ±0.08	50.0 ±0.04	<u>43.6</u> ±0.06	<u>47.6</u>	65.5 ±0.03	65.8 ±0.02	<u>59.1</u> ±0.06	<u>63.5</u>	$N_{batch} \times 2$	N_{batch}
ABLATION STUDY OF DIFFERENT SAMPLE PURIFICATION SCOPES.										
+ PURIF. ALL SAMPS.	49.5 ±0.00	49.9 ±0.10	44.6 ±0.06	48.0	65.3 ±0.06	65.8 ±0.01	59.6 ±0.01	63.5	$N_{batch} \times 2$	N_{batch}
+ PURIF. MALI. SAMPS.	49.1±0.02	49.6±0.08	44.3±0.06	47.7	64.7±0.04	65.3±0.06	59.0±0.07	63.0	$N_{batch} \times 2$	N_{batch}

Experiments

Table 2: Experimental results (top-1 classification accuracy (%)) on the lifelong TTA task.

METHODS	ROUND										AVERAGE
	R-1	R-2	R-3	R-4	R-5	R-6	R-7	R-8	R-9	R-10	
NOADAPT	31.6±0.00	31.6±0.00	31.6±0.00	31.6±0.00	31.6±0.00	31.6±0.00	31.6±0.00	31.6±0.00	31.6±0.00	31.6±0.00	31.6±0.00
TENT	8.1±0.06	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.9±0.01
+ PTTA	60.0±0.05	31.8±0.11	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	0.1±0.00	9.3±0.02
COTTA	42.9±0.25	40.6±0.57	37.0±1.33	34.8±0.82	33.5±1.02	32.0±0.67	31.0±0.50	30.5±0.61	30.7±0.89	30.6±0.90	34.4±0.53
+ PTTA	52.1±0.28	46.3±0.25	42.8±0.04	40.3±0.06	39.2±0.09	38.8±0.01	38.4±0.14	38.2±0.03	38.0±0.05	37.6±0.08	41.2±0.02
SOTTA	59.5±0.22	60.7±0.15	61.0±0.15	61.3±0.16	61.5±0.13	61.5±0.19	61.6±0.15	61.7±0.15	61.8±0.22	61.9±0.20	61.3±0.24
+ PTTA	61.3±0.39	62.6±0.42	63.1±0.34	63.4±0.31	63.5±0.27	63.7±0.26	63.8±0.27	63.9±0.25	63.9±0.25	64.0±0.24	63.3±0.43
SAR	60.0±0.02	61.1±0.02	61.4±0.02	61.6±0.02	61.7±0.02	61.8±0.03	61.7±0.02	61.6±0.08	59.3±0.26	60.4±0.06	61.1±0.04
+ PTTA	61.6±0.01	63.0±0.00	63.4±0.00	63.6±0.03	63.8±0.02	63.9±0.01	64.0±0.02	64.0±0.02	64.1±0.00	64.1±0.02	63.5±0.00
ETA	62.2±0.07	59.5±0.13	55.4±0.39	46.6±7.84	31.5±27.2	29.5±25.4	28.2±24.3	26.2±22.6	26.1±22.6	25.1±21.6	39.0±15.2
+ PTTA	65.3±0.06	65.4±0.03	65.3±0.06	65.1±0.01	64.9±0.03	64.7±0.05	64.6±0.03	64.4±0.06	64.3±0.04	64.1±0.04	64.8±0.00
EATA	62.5±0.40	62.2±0.39	61.9±0.37	61.7±0.44	61.6±0.48	61.4±0.44	61.3±0.53	61.2±0.38	61.0±0.42	61.0±0.35	61.6±0.41
+ PTTA	64.7±0.31	65.0±0.43	65.0±0.39	65.1±0.43	65.1±0.40	65.0±0.38	65.0±0.45	65.0±0.43	65.0±0.39	65.0±0.41	65.0±0.40
DEYO	62.0±0.50	43.1±18.9	32.7±29.0	25.0±27.8	17.9±30.7	16.8±28.8	15.9±27.4	3.5±5.94	0.1±0.01	0.1±0.00	21.7±15.9
+ PTTA	65.8±0.01	66.0±0.05	66.0±0.04	65.9±0.05	65.9±0.05	65.8±0.07	65.7±0.05	65.5±0.07	64.9±0.91	61.6±4.30	65.3±0.53
CPL	55.5±0.13	57.3±0.11	57.8±0.10	58.1±0.15	58.4±0.27	58.5±0.21	58.5±0.20	58.7±0.11	58.8±0.10	58.8±0.21	58.0±0.13
+ PTTA	59.7±0.02	61.5±0.02	62.0±0.03	62.3±0.02	62.6±0.02	62.7±0.00	62.9±0.03	63.0±0.00	63.1±0.01	63.2±0.03	62.3±0.01

Experiments

Table 3: Experimental results (top-1 classification accuracy (%)) on the single and continual TTA tasks for ImageNet-C (left) and other ImageNet variants (right). We highlight the highest accuracy in **bold** and the second best as underline.

METHODS	RESNET50		ViT-B/16		METHODS	IMAGENET					AVG.
	SINGLE	CONTL.	SINGLE	CONTL.		-1K	-A	-V2.	-R.	-S.	
NOADAPT	18.2±0.00	18.2±0.00	31.6±0.00	31.6±0.00	NOADAPT	78.0±0.00	23.9±0.00	66.3±0.00	43.2±0.00	18.2±0.00	45.9±0.00
TENT	42.8±0.04	9.9±0.22	51.3±0.31	14.2±10.5	TENT	82.0±0.03	29.0±0.57	70.3±0.20	17.6±0.02	2.0±0.24	40.2±0.10
+ PTTA	44.9±0.05	36.9±1.82	52.1±0.26	60.1±0.08	+ PTTA	82.0±0.07	28.7±0.75	70.4±0.29	19.1±0.62	2.5±0.25	40.5±0.83
COTTA	32.9±0.01	33.3±0.04	38.5±0.69	42.9±0.25	COTTA	79.6±0.18	26.8±0.64	68.3±0.16	53.9±0.98	13.8±9.40	48.5±1.71
+ PTTA	34.0±0.53	37.6±0.02	48.4±4.65	51.3±0.73	+ PTTA	81.0±0.00	26.9±0.66	70.1±0.12	51.1±0.27	22.3±0.45	50.3±0.25
SOTTA	41.5±0.01	30.6±0.26	57.1±0.31	59.5±0.31	SOTTA	82.2±0.09	29.6±0.50	70.4±0.04	57.4±0.12	40.8±0.30	56.1±0.11
+ PTTA	42.9±0.36	37.0±0.69	58.1±0.49	61.3±0.55	+ PTTA	82.0±0.04	30.1±0.12	70.5±0.06	57.6±0.25	40.9±0.16	56.2±0.11
SAR	44.0±0.15	40.0±0.61	56.7±0.19	60.0±0.04	SAR	81.0±0.05	30.0±0.74	70.3±0.11	43.3±0.18	23.6±1.35	49.6±0.36
+ PTTA	45.5±0.93	46.2±0.02	58.0±0.04	61.6±0.03	+ PTTA	82.1±0.03	30.5±0.23	70.5±0.17	43.2±0.02	24.3±0.55	50.1±0.19
ETA	48.2±0.04	43.7±0.06	63.7±0.03	62.2±0.07	ETA	82.3±0.05	29.8±1.19	71.3±0.22	56.5±0.90	<u>45.9±0.14</u>	57.2±0.18
+ PTTA	<u>49.5±0.00</u>	<u>48.5±0.05</u>	<u>64.4±0.04</u>	<u>65.3±0.06</u>	+ PTTA	82.7±0.02	32.7±2.47	71.7±0.06	61.4±0.04	47.4±0.09	<u>59.2±0.53</u>
EATA	48.0±0.06	48.4±0.03	61.7±0.51	62.2±0.63	EATA	81.7±0.07	31.4±0.79	70.8±0.06	57.7±0.71	44.6±0.58	57.2±0.15
+ PTTA	49.1±0.08	50.3±0.04	62.7±0.35	64.7±0.31	+ PTTA	81.9±0.07	30.8±0.82	71.1±0.11	59.9±0.33	45.6±0.56	57.9±0.08
DEYO	48.6±0.11	9.2±0.25	64.2±0.46	62.2±0.35	DEYO	82.2±0.11	<u>39.1±0.34</u>	<u>71.8±0.25</u>	59.5±0.28	40.8±1.30	58.7±0.13
+ PTTA	49.9±0.10	25.7±4.98	65.3±0.01	65.8±0.01	+ PTTA	<u>82.7±0.10</u>	40.7±0.85	71.9±0.12	<u>61.0±0.41</u>	42.4±3.18	59.7±0.57
CPL	42.8±0.04	42.3±0.15	52.4±0.08	55.5±0.13	CPL	80.7±0.17	27.7±0.07	69.3±0.27	55.1±0.11	39.5±0.17	54.5±0.07
+ PTTA	44.6±0.06	45.3±0.02	54.6±0.08	59.7±0.02	+ PTTA	81.2±0.12	27.4±0.32	69.5±0.45	56.0±0.21	39.3±0.27	54.7±0.08

Experi

Table 4: Experimental results (top-1 classification accuracy (%)) for test time prompt tuning on the episodic TTA task.

METHODS	IMAGENET					AVG.
	-1K	-A	-V2.	-R.	-S.	
CLIP-RN50						
ZERO-SHOT	58.2±0.00	21.8±0.00	51.4±0.00	56.2±0.00	33.4±0.00	44.2±0.00
TPT	58.7±0.05	21.9±0.36	52.1±0.29	57.0±0.14	33.3±0.05	44.6±0.16
+ PTTA (ID)	59.1±0.01	22.6±0.16	52.4±0.19	58.0±0.10	33.5±0.06	45.1±0.05
+ PTTA (OOD)	59.1±0.07	22.5±0.44	52.3±0.07	57.9±0.12	33.4±0.04	45.0±0.14
TPT (CoOp)	62.8±0.05	22.6±0.24	55.1±0.16	56.0±0.02	34.2±0.07	46.1±0.06
+ PTTA (ID)	62.6±0.06	23.9±0.12	55.4±0.04	57.5±0.02	34.5±0.06	46.8±0.02
+ PTTA (OOD)	62.7±0.07	23.5±0.91	55.4±0.42	57.1±1.01	34.3±0.09	46.6±0.46
CLIP-ViT-B/16						
ZERO-SHOT	66.7±0.00	47.9±0.00	60.9±0.00	74.0±0.00	46.1±0.00	59.1±0.00
TPT	66.2±0.05	48.1±0.52	59.9±0.11	74.3±0.12	45.8±0.11	58.9±0.10
+ PTTA (ID)	66.1±0.09	48.7±0.23	60.0±0.08	75.1±0.08	45.7±0.12	59.1±0.07
+ PTTA (OOD)	66.6±0.08	49.1±0.03	60.0±0.08	75.3±0.02	45.9±0.10	59.4±0.05
TPT (CoOp)	71.3±0.04	49.6±0.27	63.9±0.12	75.0±0.11	47.5±0.07	61.5±0.08
+ PTTA (ID)	71.1±0.02	49.7±0.04	64.1±0.08	73.5±0.08	47.2±0.04	61.1±0.00
+ PTTA (OOD)	71.2±0.08	50.0±0.30	63.9±0.19	75.6±0.05	47.6±0.07	61.6±0.05

Table 5: Experimental results for semantic segmentation on episodic and continual TTA tasks. We report the mIoU.

METHODS	DAY2NIGHT	CLEAR2FOG	CLEAR2RAIN	DYNAMIC	HIGHWAY
NOADAPT	58.4±0.00	52.8±0.00	71.8±0.00	46.6±0.00	28.7±0.00
COTTA	61.3±0.07	56.7±0.07	70.8±0.11	46.4±0.08	33.6±0.03
MEMO-EPI.	61.1±0.01	55.1±0.03	71.7±0.01	50.3±0.01	35.3±0.12
+ PTTA	61.4±0.01	55.3±0.02	72.1±0.01	50.9±0.01	35.3±0.11
TENT-CONTL.	61.5±0.01	56.0±0.02	70.9±0.02	50.4±0.02	32.0±0.01
+ PTTA	62.3±0.03	56.8±0.22	72.5±0.01	52.6±0.03	33.7±0.08

Table 6: Error Rate (%) of adversarial attacked samples under the indiscriminate and instant attack scenario.

METHODS	TENT	ETA	DEYO	CPL	AVG. ↓
<i>CIFAR100-C</i>					
BN ADAPT	53.3±0.16	52.4±0.24	54.2±0.09	54.9±0.02	53.7±1.06
+ PTTA	52.0±0.18	51.5±0.23	52.3±0.24	54.6±0.12	52.6±1.40
MEDBN	46.6±0.34	45.9±0.14	48.7±0.01	47.2±0.03	47.1±1.17
+ PTTA	45.7±0.30	45.3±0.02	46.6±0.10	47.1±0.21	46.2±0.82
<i>ImageNet-C</i>					
BN ADAPT	69.4±0.11	63.2±0.02	62.6±0.03	66.1±0.07	65.3±3.11
+ PTTA	65.4±0.79	62.1±0.03	62.1±0.57	64.2±0.01	63.5±1.63

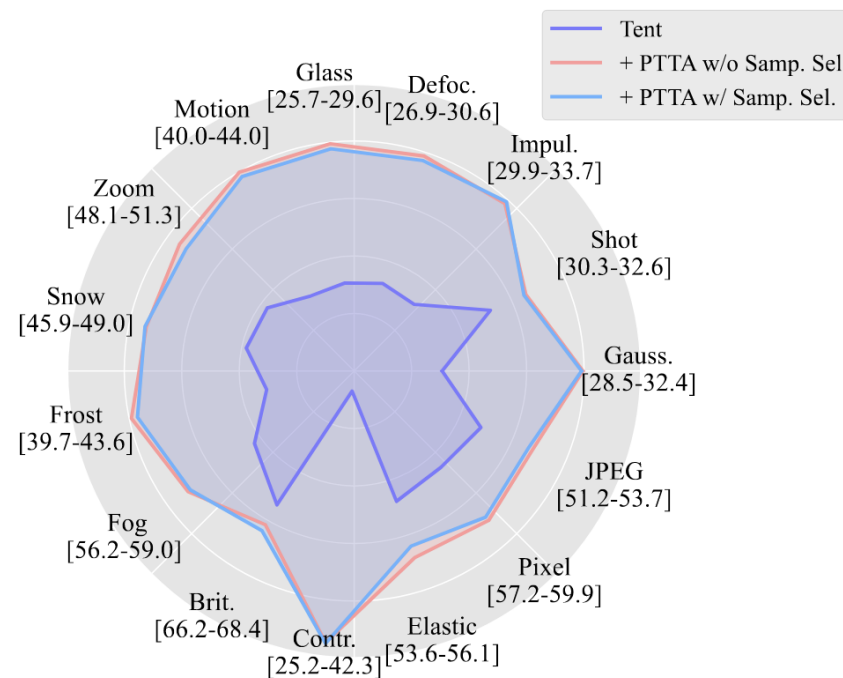


Figure 5: Comparison of selecting benign samples and using all samples as candidates for purifying malicious samples.



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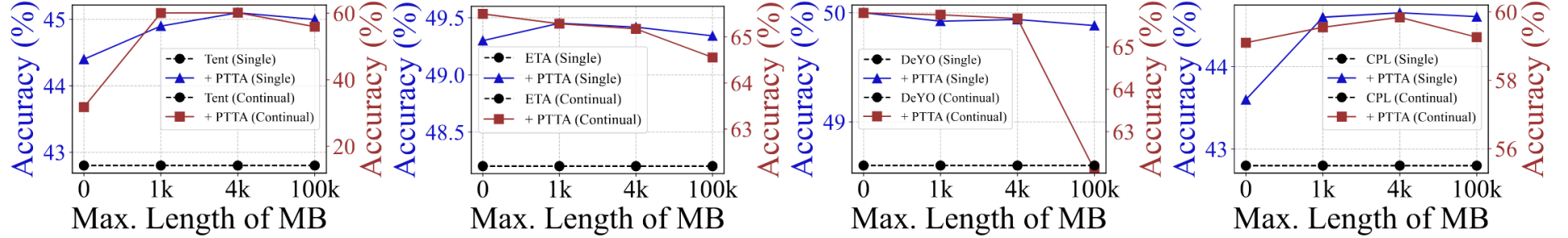


Table 7: Ablation study on retrieving top- K benign samples for malicious sample purification.

METHODS	SINGLE			
	$K = 1$	$K = 2$	$K = 3$	AVG.
TENT	42.8 $\pm$ 0.04	42.8 $\pm$ 0.04	42.8 $\pm$ 0.04	42.8 $\pm$ 0.04
+ PTTA	44.9 $\pm$ 0.05	44.7 $\pm$ 0.04	44.5 $\pm$ 0.03	44.7 $\pm$ 0.22
ETA	48.2 $\pm$ 0.04	48.2 $\pm$ 0.04	48.2 $\pm$ 0.04	48.2 $\pm$ 0.04
+ PTTA	49.0 $\pm$ 0.05	49.1 $\pm$ 0.09	49.1 $\pm$ 0.06	49.1 $\pm$ 0.06
DeYO	48.6 $\pm$ 0.11	48.6 $\pm$ 0.11	48.6 $\pm$ 0.11	48.6 $\pm$ 0.11
+ PTTA	49.3 $\pm$ 0.06	49.3 $\pm$ 0.16	49.4 $\pm$ 0.00	49.3 $\pm$ 0.03
CPL	42.8 $\pm$ 0.04	42.8 $\pm$ 0.04	42.8 $\pm$ 0.04	42.8 $\pm$ 0.04
+ PTTA	44.6 $\pm$ 0.06	44.4 $\pm$ 0.05	44.1 $\pm$ 0.08	44.4 $\pm$ 0.25

Table 8: Ablation study on retrieving top- K benign samples for malicious sample purification (Continual)

METHODS	CONTINUAL			
	$K = 1$	$K = 2$	$K = 3$	AVG.
TENT	14.2 $\pm$ 10.5	14.2 $\pm$ 10.5	14.2 $\pm$ 10.5	14.2 $\pm$ 10.5
+ PTTA	60.1 $\pm$ 0.08	59.7 $\pm$ 0.01	59.4 $\pm$ 0.05	59.7 $\pm$ 0.32
ETA	62.2 $\pm$ 0.07	62.2 $\pm$ 0.07	62.2 $\pm$ 0.07	62.2 $\pm$ 0.07
+ PTTA	64.7 $\pm$ 0.04	64.3 $\pm$ 0.07	63.9 $\pm$ 0.07	64.2 $\pm$ 0.43
DeYO	62.2 $\pm$ 0.35	62.2 $\pm$ 0.35	62.2 $\pm$ 0.35	62.2 $\pm$ 0.35
+ PTTA	65.1 $\pm$ 0.03	64.9 $\pm$ 0.01	64.4 $\pm$ 0.32	64.9 $\pm$ 0.21
CPL	55.5 $\pm$ 0.13	55.5 $\pm$ 0.13	55.5 $\pm$ 0.13	55.5 $\pm$ 0.13
+ PTTA	59.6 $\pm$ 0.01	59.1 $\pm$ 0.02	58.4 $\pm$ 0.04	59.0 $\pm$ 0.58

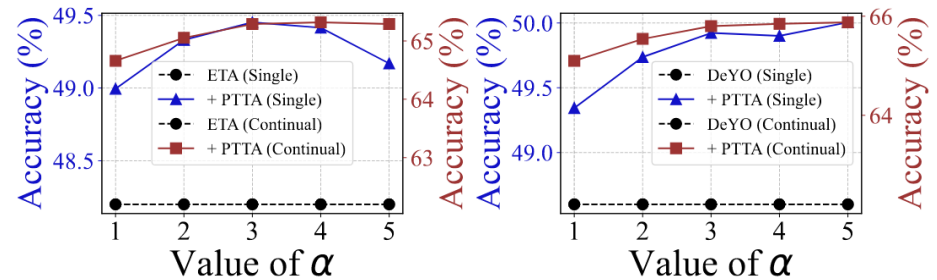


Figure 8: Ablation Study of α in Eq. 8.

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Table 18: The comparison of PTTA with the vanilla Mixup method. Also the comparison of different purification scopes in PTTA, which purify all test samples by default.

METHODS	SINGLE				CONTINUAL			
	ETA	DEYO	CPL	AVG.	ETA	DEYO	CPL	AVG.
BASE ALGORITHM	48.2	48.6	42.8	46.5	62.2	62.2	55.5	60.0
+ VANILLA MIXUP (ZHANG ET AL., 2018) (w/ 1K MEMORY BANK)	48.6	49.1	38.2	45.3	64.7	64.5	55.9	61.7
+ PURIFY ONLY BENIGN SAMPLES (w/ OOD RETRIEVAL)	48.9	<u>49.6</u>	<u>44.4</u>	47.6	<u>65.2</u>	<u>65.3</u>	<u>59.5</u>	<u>63.3</u>
+ PURIFY ONLY MALICIOUS SAMPLES (w/ OOD RETRIEVAL)	<u>49.1</u>	49.6	44.3	<u>47.7</u>	64.7	65.3	59.0	63.0
+ PTTA (w/ OOD RETRIEVAL)	49.5	49.9	44.6	48.0	65.3	65.8	59.6	63.5

Thank You!